

Experiment 01: Systems Boundary Analysis

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1. Objective

To write and analyze a C program executing low-level system operations, calculating the asymptotic time complexity $O(n^2)$ of the resulting buffer allocation.

2. Implementation Code

```
#include <stdio.h>

int main() {
    int buffer[5] = {1, 2, 3, 4, 5};
    for(int i = 0; i < 5; i++) {
        printf("Memory Address: %p\n", (void*)&buffer[i]);
    }
    return 0;
}
```